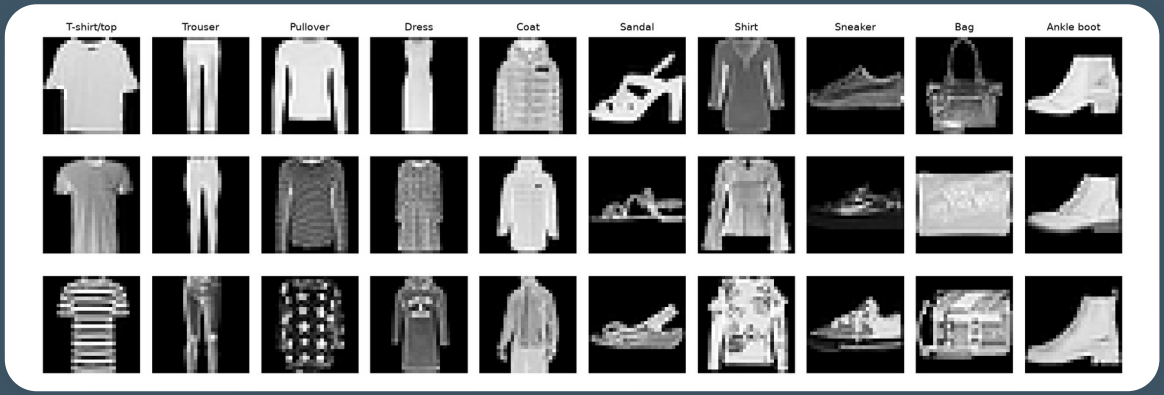
PROJECT 11 · PHASE 1

Inventory Image Tagging

Teaching simple classifiers to tell clothes apart



10 clothing categories

28 × 28 grayscale images

8 models compared

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The problem we set out to solve

A shop has thousands of clothing photos. Given one image, which of ten categories is it?

1

Compare two ways of seeing an image

Raw pixel brightness versus HOG edge directions, each tried with four classifiers.

2

Test it fairly

Choose settings on validation images only. Freeze choices before evaluating the official test set.

3

Explain the mistakes

Find out which clothes get confused, especially shirts, T-shirts and pullovers.

What made it hard

28 × 28

Tiny pictures

Only 784 gray values per image. No color and very little fine detail to lean on.

59.2%

Lookalike tops

Shirts get mixed up with T-shirts, pullovers and coats. They have the lowest recall of all ten.

Fixed

Honest testing

Settings come from 12,000 validation images. Freeze choices before the final 10,000-image test.

299.5 s

A slow linear SVM

An early fit took five minutes. We adjusted scaling and stopping settings before testing.

How we tackled it

1. **Fashion-MNIST** → 2. **Shared split** → 3. **Pixels or HOG** → 4. **4 classifiers** → 5. **Pick on validation**

48,000
images to train every model

12,000
images to choose settings

10,000
official images for the final test

Same images for all 8 combinations (2 representations × 4 classifiers), so the comparison stays fair.

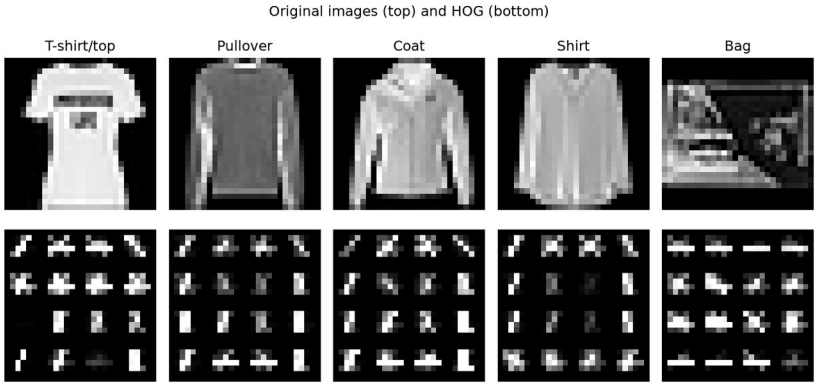
Two ways to describe a picture

Pixels **784 values**

How bright each pixel is. Keeps exactly where things sit in the image.

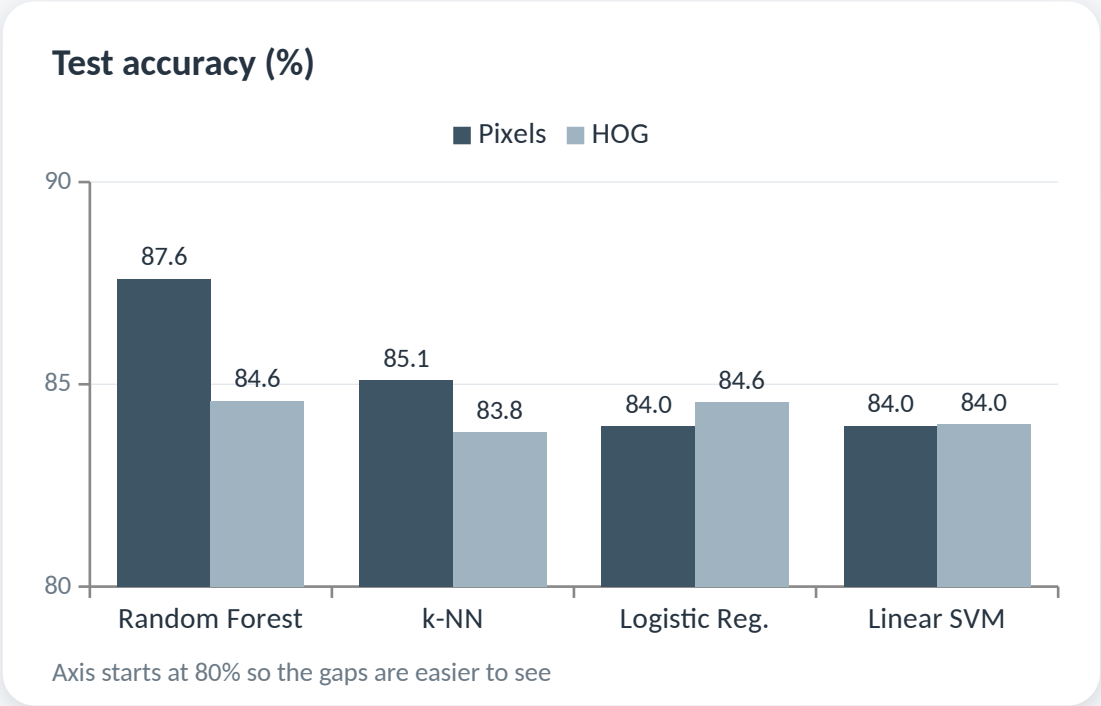
HOG **324 values**

Which way edges point in small patches. Cares about shape more than lighting.



Top row: the original 28 × 28 images. Bottom row: what HOG sees. Shirts and pullovers still look alike.

Best of our eight models



Axis starts at 80% so the gaps are easier to see

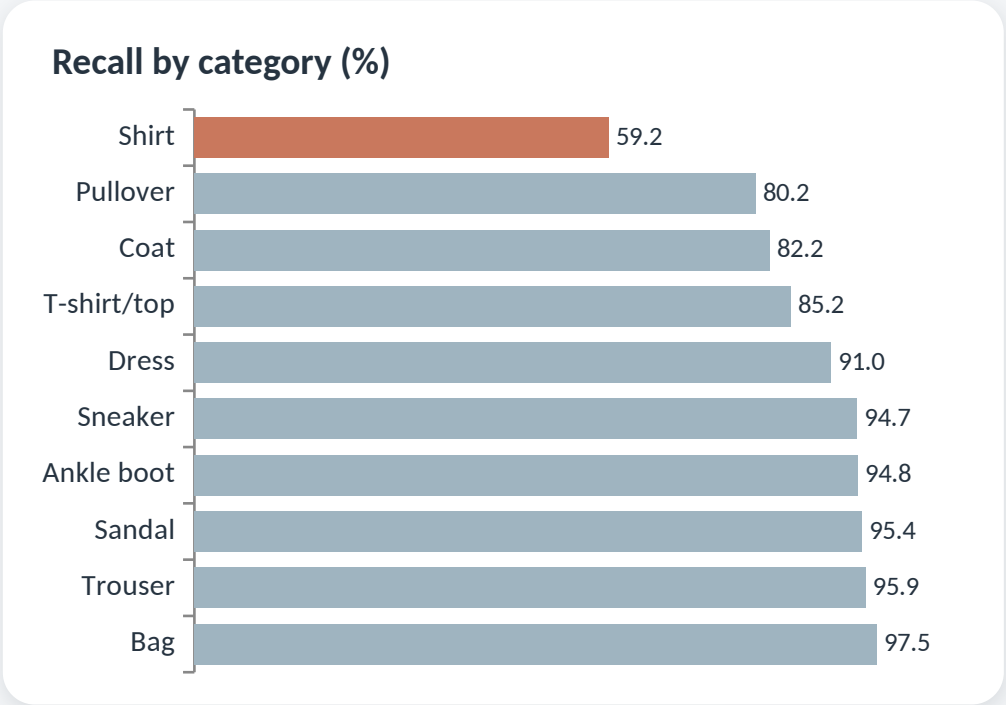
Picked on validation

87.61%

of test images tagged correctly by Pixels + Random Forest

0.8747	macro F1
24.98 ms	median prediction
69.96 MiB	saved model

Where it gets confused



The most common mix-ups

149 shirts called T-shirt/top

124 shirts called Pullover

116 pullovers called Coat

Try it in the dashboard

1

Pick an image

Use a built-in sample or upload your own clothing photo.

2

Choose a model

Default is Pixels + Random Forest. Switch it in settings.

3

Quick preview

See the 28 × 28 version the model actually looks at.

4

Get a prediction

The category shows up with the prediction time.

inventory-image-tagging.streamlit.app

Ordinary photos are experimental. Only the 28 × 28 benchmark images have measured accuracy.

What we learned, what's next

What we learned

1. Pixels + Random Forest led the eight: 87.61% accuracy, macro F1 0.8747.
2. HOG helped Logistic Regression and Linear SVM a little, and hurt k-NN and Random Forest.
3. Shirts stay the hardest category.

Still open

1. We used one split and one seed, with a limited search.
2. Ordinary photos, near-duplicates and new categories are untested.
3. Phase 2: a small CNN on the same split.